Product → Engineering: OK, thanks—sounds good. I will work with Sales on the details.

Product → Sales: Alright, we can commit to 99% availability over the time period of a month for the Chat2Much API, and there is some wiggle room on our end already baked in with it. We are using SLIs `A`*,* `B`*, and* `C` *to track this SLO.*

Sales → Product: Perfect, thanks! I will use this and attach the financial implications—that is, the credits that a customer receives in the case we fail to meet the SLO.

TIP To arrive at an availability percentage, you can work backward from how many errors you want to tolerate at most. Let's say you've observed that your service in load testing has received 1,000,000 requests in a period of time and that in the same time period, there were 6,942 failed requests. To implement some headroom, you can round up, so make this 7,000. Therefore, you can commit to 99% availability, realistically.

Continuing the example, the Chat2Much API then goes into production, and engineering is using dashboards and alerts to track the SLIs `A`, `B`, and `C`. They report on a monthly basis how the SLO is met and if not, by how much. The customer knows the availability from the SLA and can see how the API is performing against the contractually agreed upon goals on their own dashboards. Everyone is happy—end of story.

If you want to dive deeper into this topic, I recommend the following resources:

- *Implementing Service Level Objectives* by Alex Hidalgo (2020, O'Reilly Addison-Wesley, http://mng.bz/5w5q)
- "From Critical User Journey to SLO/SLIs" by Adam Roberts (http://mng.bz/6D65)
- "The SLO Development Lifecycle" (SLODLC; https://www .slodlc.com/)
- "Effective SLOs and You" (http://mng.bz/o1zj)
- "A Practical Guide for Implementing SLO" by Prathamesh Sonpatki and Saurabh Hirani (http://mng.bz/nWG8)

Let's now shift gears and look at a concrete domain: Prometheus. We'll also look at how to implement SLOs in the context of this ecosystem.

10.2.2 Using Prometheus to implement SLOs

After our theoretical treatment on what SLAs, SLOs, and SLIs are and how to implement them conceptually, we will now have a closer look at how to specifically implement SLOs and SLIs with open source tooling from and in the Prometheus ecosystem. As with all other chapters in the book, the idea is that you can try out and evaluate things without costs or (license) commitments involved, and the steps should be applicable more or less to any provider in this space.

Pyrra

Pyrra (https://github.com/pyrra-dev/pyrra) is an open source tool that aims to make SLOs with Prometheus easy to use. Pyrra focuses on Prometheus as the source of the